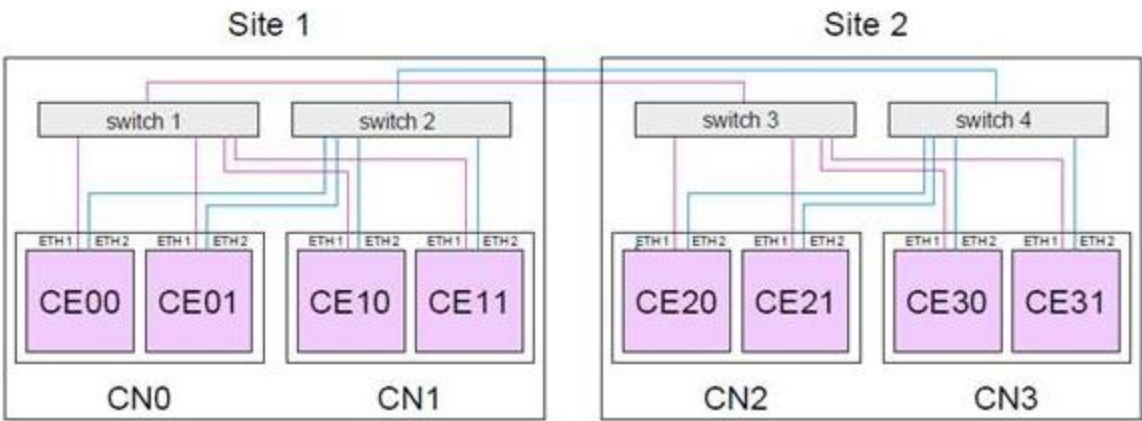


Split Brain

SSS_VIL Redundancy & Architecture Requirements 3BU 1500 6955 DTAPA	Synopsis	Origin	Allocation
EL4-VIL-SSS-REQ-00222 VIL Split-Brain Prevention	It shall be ensured that only one subgroup of synchronized nodes remains active when VIL nodes are disconnected, preventing the Split-Brain problem (critical for availability),	[EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy	VIL ELEKTRA Core ECMUX

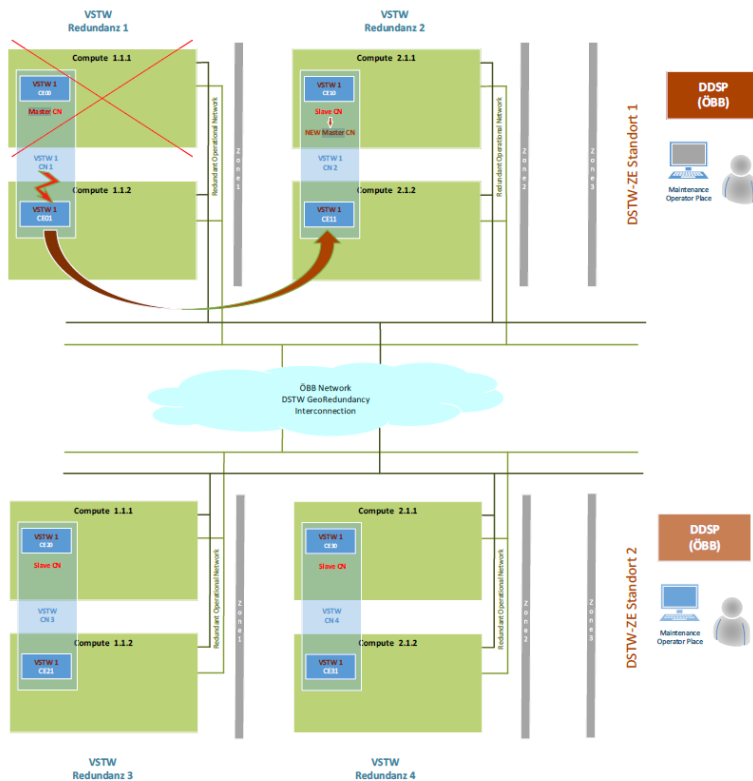


The **2(4)×2oo2 HotHot redundant architecture** of the TAS Platform differentiates between one **Master Computing Node (Master CN)** and several **Slave Computing Nodes (Slave CNs)**.

The **Master CN** has a central and critical role because it controls the **synchronous redundant system**. For the system to remain operational, **exactly one CN must act as Master at all times**.

If one hardware instance (one CE — Computing Element) of the Master CN fails, the **remaining CE** of that Master CN will—before it also becomes inactive—**inform all Slave CNs about the failure of the Master CN**.

After receiving this information, the **remaining Slave CNs automatically elect a new Master CN**. This process is fully automatic and does **not** interrupt operation.



A simultaneous failure of both servers (CEs) of the Master Computing Node (Master CN)

This situation may occur, for example, through a **complete and immediate power loss affecting both Master CN servers at the same time**.

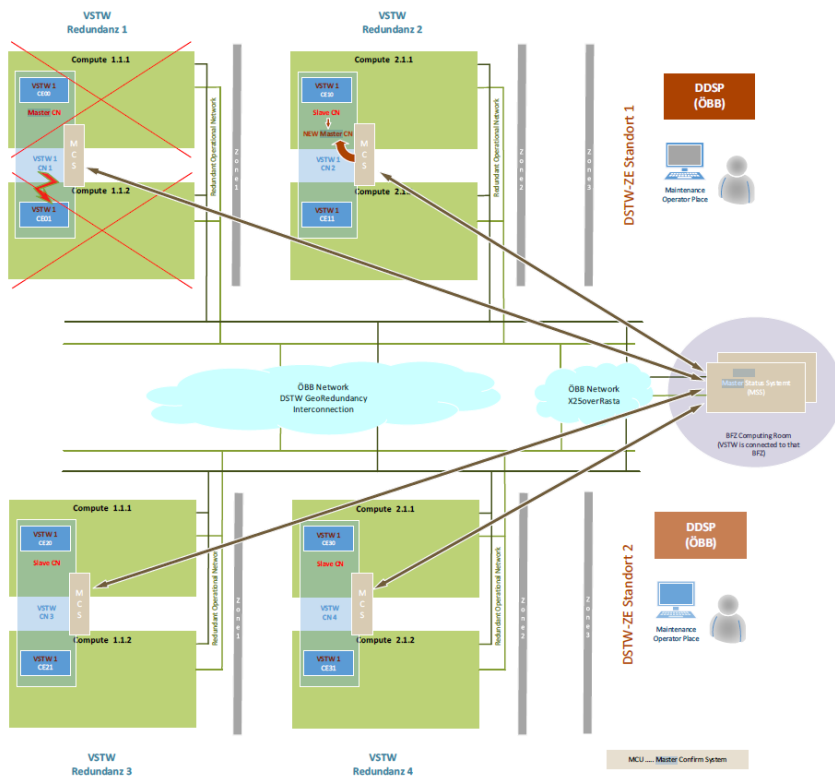
In this scenario, **neither of the two Master CEs is able to inform the Slave CNs about the Master failure**.

To maintain system safety, the architecture must **strictly avoid a situation where two Master CNs exist simultaneously** (SplitBrain). To safely handle this doublefailure scenario, the system must query an **independent third instance** before electing a new Master.

To ensure that the Master decision is not influenced by local network problems, the system uses a **second redundant communication path** connecting to a dedicated decision instance.

1. MSS – Master Status System (central, redundant)
 - Collects the **status of all CNs** continuously.
 - Should be located at an **independent site**, unaffected by local DSTW issues.
 - Preferably installed in the **related BFZ environment** and connected via the **X25overRaSTA network**.
 - Maintains continuous communication with every VSTW instance (all CNs).
 - Sends the collected state information back to the CNs.
2. MCS – Master Confirm System (local at each site)
 - Receives the status information from the MSS.
 - **Decides locally** whether a Slave CN is allowed to become the new Master.
 - Since the MSS data arrives through **alternative network paths**, the MCS can reliably determine whether the Master CN truly failed or is simply unreachable.

Even if **both Master CN servers fail simultaneously**, the system can **still safely and uniquely determine a new Master CN**.



Update 03 Mar 2026

V:\ORG\RL\Produkte\06.02_ELEKTRA\06.3_Devel\08_DSTW_ELEKTRA_4.0\02_RAM\08_SplitBrain

Split-brain analysis and proposals for DIL-CU states:

Summary and decision to be made

Two links architecture:

- Works in 2x2oo2 setup
- In 4x2oo2 there is a potential for split-brain (and safety hazard) in case of double failures on the connection between geo-site.
 - **What is the probability of such situation? What is availability of such network?**
 - **Is it within the residual risk?**

RAM:

Prelim_RAM_Estimation_DSTW_Ed02_20260303.xls

Preliminary calculations of the network give the following availability values:

	Components	FiT	MTBF y	Availability %	sec year	MTTR
Site 1	Network = 3*(Switch1oo2)	4	25 666	99.999 996 4	1.1	8 h
	Core_Router = Router1oo2	1	76 999	99.999 998 8	0.4	8 h
	Total	6	19 250	99.999 995 3	1.5	8 h
Site 2	Network = 3*(Switch1oo2)	4	25 666	99.999 996 4	1.1	8 h
	Core_Router = Router1oo2	1	76 999	99.999 998 8	0.4	8 h
	Total	6	19 250	99.999 995 3	1.5	8 h

The use of *CommonCause Failure (CCF)* analysis and the *factor* methodology from **IEC615086** is applied following the tables D.1 to D.6 of how to determine a -factor.

Table D.4 – Calculation of β_{int} or $\beta_{D int}$

Score (S or S_D)	Corresponding value of β_{int} or $\beta_{D int}$ for the:	
	Logic subsystem	Sensors or final elements
120 or above	0,5 %	1 %
70 to 120	1 %	2 %
45 to 70	2 %	5 %
Less than 45	5 %	10 %
NOTE 1 The maximum levels of $\beta_{D int}$ shown in this table are lower than would normally be used, reflecting the use of the techniques specified elsewhere in this standard for the reduction in the probability of systematic failures as a whole, and of common cause failures as a result of this.		
NOTE 2 Values of $\beta_{D int}$ lower than 0,5 % for the logic subsystem and 1 % for the sensors would be difficult to justify.		

Switches and Router are "Sensors or final elements" and we consider a Score of 120 or above which stands for a diverse system with good diagnostic testing (Table D.6) factor = 1% (0.01).

As comparison, for a non diverse system with poor diagnostic testing (Table D.6) factor = 10% (0.1).

factor	Geo-redundant Connection	FiT	MTBF y	Availability %	sec year
0	Site 1 + Site 2	11.86	9 625	99.999 990 5	2.99
0.01	Site 1 + Site 2	11.98	9 530	99.999 990 4	3.02
0.1	Site 1 + Site 2	13.05	8 750	99.999 989 6	3.29

The "worst case" scenario with a factor = 10% is listed her as an example, that the required availability of 99.999 9% (downtime 31.53 sec/year) is fulfilled even in this case.

From RAM point of view, a factor of no more than 5% should be applied.

The required network availability is clearly fulfilled.

Availability Requirements:

EL4-DIL-SSS-REQ-00307	Availability %	sec year
The network availability of a [DIL] location shall be at least	99.999 9	31.53

Meeting 13.03.2026

The ÖBB's Digital Interlocking (DIL) is designed as a geographically redundant system with two physically separated datacenter locations (geosites). The connection between the sites is provided via two redundant geolinks.

A splitbrain condition occurs when both geolinks fail simultaneously while the internal observer channel is still available: in this situation, both sites select their own master and communicate independently with the external equipment — a state classified as a safety hazard

Class	Availability	Unavailability	min/year	Subsystem
1	99.999	$1 \cdot 10^{-5}$	5,3	DIL-CU, IPS
2	99.995	$5 \cdot 10^{-5}$	26.3	VIL, FES
3	99.99	$1 \cdot 10^{-4}$	52.6	
Network	99.999 9	$1 \cdot 10^{-6}$	0.5	(REQ-00307) = internal Network

The availability requirements in the SSS RAMS (REQ00406ff.) apply to the subsystems supplied by Hitachi Rail: DILCU, VIL, IPS, FES, OC. The geolinks are **external ÖBB infrastructure** and are therefore not subject to these requirements.

- **Step 1:** IPS Availability Overall:

REQ-01480: Availability — IPS — Overall: The IPS shall have a class 1 overall availability of at least 99.999%

this leads to an Unavailability of $U_{IPS} = 1 - A_{IPS} = 1 - 0,99999 = 1 \times 10^{-5}$

- **Step 2:** Internal Network

REQ-00307: The network availability of a DIL location shall be at least 99.9999%

only for the internal network at one site, leads to : $U_{network} = 1 - A_{network} = 1 - 0,999999 = 1 \times 10^{-6}$

- **Step 3:** CCF

The two geolinks run along physically separate cable routes, but both are operated by ÖBB. This results in a common organisational failure mode (CCF). In accordance with IEC615086 AnnexD, a CCF factor is assessed.

= 0.02

- **Step 4:** Allocation to IPS Unavailability

Unavailability-Budget of IPS ($U_{IPS} = 10^{-5}$)

We say "The geolink dual failure (splitbrain exposure) is one of several failure modes and An allocation share of = 10% is assigned to the geolink dual failure." (very conservative estimation)

$P_{Split-Brain} = U_{IPS} \times 0,10 = 1 \times 10^{-5} \times 0,10 = 1 \times 10^{-6}$

- **Step 5:** Availability estimation of geo-link

$P_{Split-Brain} = 10^{-6}$ and = 0.02

$U_{Link} = 1 \times 10^{-6} / 0.02 = 5 \times 10^{-5}$

$A_{Link} = 1 - 5 \times 10^{-5} = 0.99995 = 99.995\% = \text{Class 2 Availability}$

	Availability	Unavailability	min/year	
U_{IPS}	99.999	1×10^{-5}	5.26	IPS can fail in total
U_{Link}	99.995	5×10^{-5}	26.3	One Link can fail in total
$P_{Split-Brain}$	99.999 9	10^{-6}	0.5	Both links fail

factor of 0.02 states that: only 2% of link failures will affect the second link at the same time all other 98% affect only one link = no Split Brain

$1 \times 10^{-6} \times 8760h/year = 0.00876h/year \quad 31.5 \text{ sec/year}$

IPS Unavailability budget = 5.26 min/year we allocate = 10% to the Split Brain failure.

$P_{Split-Brain} = 0.10 \times 5.26min = 31.5 \text{ sec/year}$

The IPS unavailability budget is $U_{IPS} = 10$ (equivalent to 5.26 minutes per year).

The splitbrain exposure for one line: $A_{Link} = 99.995\%$ amounts to 31 seconds per year — which corresponds to approximately 10% of the IPS budget.

Update 18.03.2026

- If 4 lines instead of 2 between the sites are used only a failure of all 4 lines leads to Split Brain from RAM point of view, this is negligible and does not need to be considered.
- Only remaining possibility of Split Brain is if both Core-Routers on one site fail at the same time Double Fault see [Split-brain analysis and proposals for DIL-CU](#)

A double fault occurs when two **independent** failures happen simultaneously or within a short period, leading to a system failure. These faults are typically considered independent and not caused by a shared factor.

In a redundant system with two components, a double fault would occur if both components fail independently due to separate causes, such as one failing due to a manufacturing defect and the other due to an unrelated environmental factor.

Assumption: We consider two Core_Router at one site, in a 1oo2_hot_redundant configuration. FIT = 10.000 per router

the following formula applies:

- Failure rate for a 1oo2 System

$$\lambda_{sys} \approx 2 \frac{\lambda^2}{\mu}$$

Double failure occurs when one router fails, and before it is repaired (within MTTR), the second router fails independently.

Assuming FIT = 10.000 per router and depending on the MTTR (1h-72h), the system failure rate sys remains reasonable low, confirming that a double failure is **not a dominant contributor**.

MTTR [h]	_sys	MTBF [y]	Availability	min/year down
1	0.20	570 776	99.999 999 98	0.0001
4	0.80	142 694	99.999 999 68	0.0017
8	1.60	71 347	99.999 998 72	0.0067
12	2.40	47 565	99.999 997 12	0.0151
24	4.80	23 782	99.999 988 48	0.0605
72	14.40	7 927	99.999 896 32	0.5449

Double failure of Core Routers does not contribute significantly to the failure of the system.

By contrast, the contribution of **CommonCause Failures (CCF)**—modeled via the **factor** and is more significant.

With a conservative $\lambda = 0.02$ and **10,000FIT per router**, the CCF path contributes **200FIT**, which outweighs the doublefailure contribution.

This indicates that the **overall system failure rate is primarily dominated by CCF events**, not by independent double failures during the repair window.

Conclusion:

- reducing MTTR reduces the doublefailure rate and the associated minutes of unavailability per year.
- Even MTTR = 72h produces only **0.54 minutes of downtime per year** from double failures.
- High Availability double faults is not a dominant contributor.
- CCF contribution exceeds the doublefailure rate across all MTTR variants.

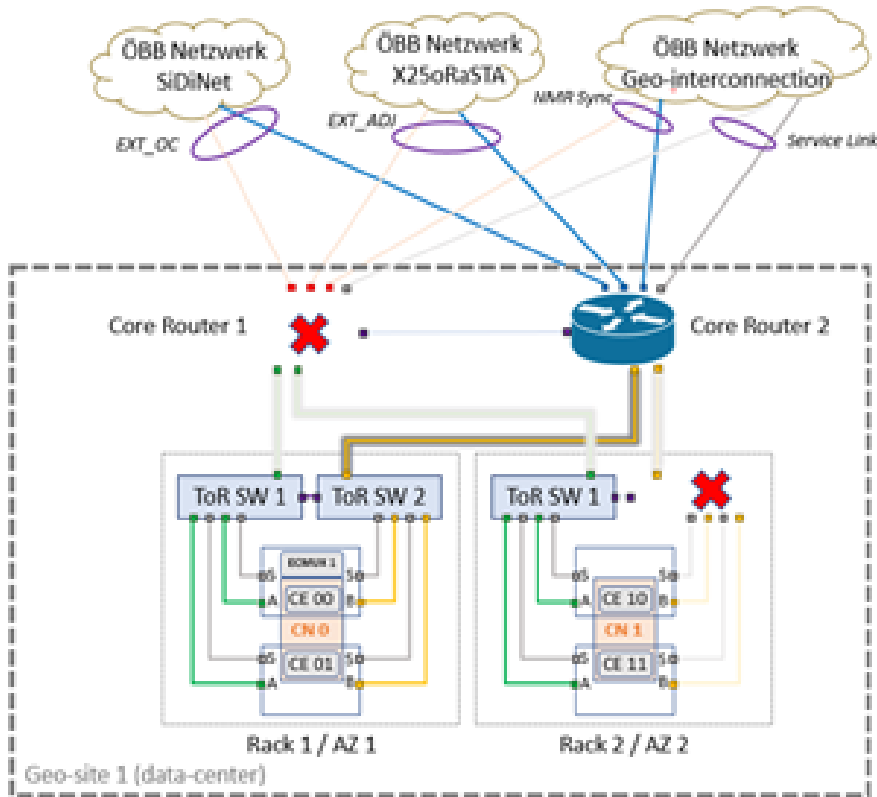
Calculations available here: [Prelim_RAM_Estimation_DSTW_Ed02_20260319.xlsx](#)

Update 19.07.2026

Context:

Assessment of the double failure “one Core Router + one ToR switch” (crossed configuration) as a split-brain precondition. The architecture is fixed: fully 1oo2 redundant — both ToR switches and Core Routers are duplicated per rack. There is no single-ToR or single-router variant.

Request: Double failure Core Router 1 – ToR SW1 (Rack 2)



My assumption, based on the figure you provided is: each cabinet has exactly two serial communication paths:

- Path A = ToR SW 1 Core Router 1
- Path B = ToR SW 2 Core Router 2

Each cabinet has two serial paths: Path A = ToR SW 1 CR 1 and Path B = ToR SW 2 CR 2. Because of the 1oo2 redundancy, a **single** fault never disables a CN — the second path always remains available.

Only the **crossed double fault** (CR 1 + ToR SW 2, or CR 2 + ToR SW 1, in the same cabinet) disables both paths simultaneously the affected CN is disconnected split-brain precondition.

This yields 2 critical combinations per cabinet × 3 cabinets = **6 critical combinations** per data center.

The two failed components belong to two nominally separate 1oo2 groups (Core Router 1oo2 and ToR SW 1oo2), both redundant paths of the same redundancy function (network attachment of the CN) are lost. It therefore lies outside the single-fault guarantee of EL4-DIL-SSS-REQ-00316.

For this double fault, we will not use the Common Cause Failure (CCF) between ToR and CR (IEC 61508-6 Annex D: different device types, locations, power; no common failure mechanism)

	FiT		MTBF y
ToR switch (Nokia 7250 IXR-e)	7000	7.0×10^{-6}	16.3
Core Router	9600	9.6×10^{-6}	11.9

One cabinet (CR1 + ToR SW2) - one combination

MTTR [h]	FiT_sys	MTBF [y]	Availability	min year down
1	0.13	849 369	99.999999866%	0.0001
4	0.54	212 342	99.999997850%	0.0011
8	1.08	106 171	99.999991398%	0.0045
12	1.61	70 781	99.999980646%	0.0102
24	3.23	35 390	99.999922586%	0.0407

All cabinets of one data center - 6 combinations

MTTR [h]	FiT_sys	MTBF [y]	Availability	min year down
1	0.81	141 562	99.9999999194%	0.0004
4	3.23	35 390	99.9999987098%	0.0068
8	6.45	17 695	99.9999948390%	0.0271
12	9.68	11 797	99.9999883878%	0.0610
24	19.35	5 898	99.9999535514%	0.2441

All cabinets of two data centers - 12 combinations

MTTR [h]	FiT_sys	MTBF [y]	Availability	min year down
1	1.61	70 781	99.9999998387%	0.0008
4	6.45	17 695	99.9999974195%	0.0136
8	12.90	8 848	99.9999896781%	0.0543
12	19.35	5 898	99.9999767757%	0.1221
24	38.71	2 949	99.9999071028%	0.4883

Same conclusion from RAM pov as with the original split brain calculations:

"Depending on the MTTR (1h-24h), the system failure rate sys remains reasonable low, confirming that a double failure is not a dominant contributor to overall system risk, additionally we do not include CCFs in this double fault, therefore its even lower for this combination"

Impact does NOT scale: each event is still one CN isolated within one site.

Excel with the calculations can be found here: [Prelim_RAM_Estimation_DSTW_Ed02_20260718.xlsx](#)